Auteur: Oubayy Ahale
Studierichting: Informatica
Oefeningenreeks 5D nr. 14

$y^2 = x^4 + x^5$

$\Leftrightarrow y = \sqrt{x^4 + x^5}$

$\Leftrightarrow y = x^2\sqrt{x+1}$

$$\int_{-1}^{0} x^2\sqrt{x+1}dx$$

$t = x + 1 \Leftrightarrow x = t - 1$

$x = -1 \ \Rightarrow \ t = 0, \, x = 0 \ \Rightarrow \ t = 1$

$$= \int_0^1 (t-1)^2 \cdot \sqrt{t}dt$$

$$= \int_0^1 \left(t^2 - 2t + 1\right) \cdot \sqrt{t}dt$$

$$= \int_0^1 \left(t^{\frac{5}{2}} - 2t^{\frac{3}{2}} + \sqrt{t}\right) dt$$

$$= \int_0^1 t^{\frac{5}{2}} dt - \int_0^1 2t^{\frac{3}{2}} dt + \int_0^1 \sqrt{t}dt$$

$$= \left[\frac{2t^{\frac{7}{2}}}{7}\right]_0^1 - \left[\frac{4t^{\frac{5}{2}}}{5}\right]_0^1 + \left[\frac{2t^{\frac{3}{2}}}{3}\right]_0^1$$

$$= \frac{16}{105}$$

Voor de volledige oppervlakte vermenigvuldigen we met 2

$$= \frac{32}{105}$$

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